Riyazi Online (Unit 1 MCQS)

1. The value of i^2 is:
   (a) i
   (b) 1
   (c) -1
   (d) 0

2. What is the sum of $i + i^2 + i^3 + i^4$?
   (a) 0
   (b) 2
   (c) 1
   (d) -1

3. Which of the following is NOT true about i:
   (a) $i^2 = -1$
   (b) $i^3 = 1$
   (c) $i^4 = 1$
   (d) $i^5 = i$

4. Find the value of i^{-3}.
   (a) i
   (b) $-i$
   (c) $1/i^3$
   (d) -1

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5. Sum of imaginary powers $i^6 + i^7 + i^8 + i^9$:
   (a) 0
   (b) $2i$
   (c) $-2i$
   (d) 1

2. Complex Numbers, Algebra, and Equality

1. If $z = 2 + 3i$, then $\bar{z} =?$
   (a) $2 - 3i$
   (b) $2 + 3i$
   (c) $3 + 2i$
   (d) $-2 - 3i$

2. The product of two complex numbers $(a + bi)(a - bi)$ is:
   (a) $a^2 + b^2$
   (b) $a^2 - b^2$
   (c) ab
   (d) $a^2 + b^2 i^2$

3. The complex numbers are equal if and only if:
   (a) Their real parts are equal
   (b) Their imaginary parts are equal
   (c) Both real and imaginary parts are equal
   (d) They lie on the same circle

4. $(2 + 3i)^2 =?$
   (a) $-5 + 12i$
   (b) $-5 - 12i$

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(c) $13 + 6i$

(d) $4 + 9i$

5. Which of the following is the additive inverse of $3 - 4i$?

(a) $3 + 4i$

(b) $-3 + 4i$

(c) $-3 - 4i$

(d) $3 - 4i$

6. Which is true for the product $z\bar{z}$?

(a) Always negative

(b) Always imaginary

(c) Always real and non-negative

(d) Always equal to 1

7. Which of the following is the correct periodicity of powers of i?

(a) 3

(b) 2

(c) 4

(d) 5

8. What is the sum of $i^2 + i^3 + i^4 + i^5$?

(a) 0

(b) -1

(c) 1

(d) 2

9. Complex conjugate of $z = x + yi$ is:

(a) $x - yi$

(b) $x + yi$

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(c) $-x - yi$

(d) $-x + yi$

10. Multiplicative inverse of $z = 1 + i$ is:

(a) $1 - i$

(b) $\frac{1-i}{2}$

(c) $-1 - i$

(d) $\frac{1+i}{2}$

11. If $z = 4 + 3i$, then $| z |^2 = ?$

(a) 25

(b) 12

(c) 5

(d) 7

12. If $z = x + iy$, and $\bar{z} = x - iy$, then $z + \bar{z} = ?$

(a) $2x$

(b) $2y$

(c) 0

(d) x

3. Modulus and Square Root of Complex Numbers

1. If $z = 3 + 4i$, then $| z | = ?$

(a) 7

(b) 5

(c) 25

(d) 1

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2. The modulus of $z = a + bi$ is:
   (a) $a^2 + b^2$
   (b) $\sqrt{a^2 + b^2}$
   (c) $| a + b |$
   (d) $| a - b |$

3. If $z_1 = 1 + i, z_2 = 1 - i$, then $| z_1 |=| z_2 |=$
   (a) 1
   (b) $\sqrt{2}$
   (c) 2
   (d) $\frac{1}{\sqrt{2}}$

4. Which of the following is always true?
   (a) $| z |=| \bar{z} |$
   (b) $| z |= -\bar{z}$
   (c) $z = \bar{z}$
   (d) $z = -\bar{z}$

4. Cube Roots of Unity

1. A cube root of unity does not include:
   (a) 0
   (b) ω
   (c) ω^2
   (d) 1

2. Roots of $z^3 = 1$ are:
   (a) $1, -i, i$
   (b) $1, \omega, \omega^2$

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(c) $1,0,-1$

(d) $-1,1,1$

3. How many 4th roots does any non-zero complex number have?

   (a) 1

   (b) 2

   (c) 4

   (d) Infinite

4. One of the cube roots of unity is:

   (a) $\cos\ 120° + i\sin\ 120°$

   (b) i

   (c) 0

   (d) -1

5. Complex Polynomial and Their Roots

1. A polynomial of degree n has how many complex roots (including multiplicity)?

   (a) n

   (b) $n-1$

   (c) $n+1$

   (d) n^2

2. Roots of $z^2+1=0$ are:

   (a) 1 and -1

   (b) i and $-i$

   (c) 0 and 1

   (d) $-i$ and 0

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3. Complex conjugate roots occur:
   (a) Randomly
   (b) In triplets
   (c) In pairs
   (d) Only on real axis

4. Factor of $z^2 - 2z + 2$ is:
   (a) $(z - 1 + i)(z - 1 - i)$
   (b) $(z - 2)(z + 1)$
   (c) $(z-1)^2$
   (d) $(z+1)^2$

5. Discriminant of quadratic $ax^2 + bx + c$ helps in:
   (a) Finding modulus
   (b) Determining nature of roots
   (c) Graphing
   (d) Polar conversion

6. Complex Numbers on Real and Imaginary Axis

1. Complex number on real axis means:
   (a) Imaginary part = 0
   (b) Real part = 0
   (c) Modulus = 0
   (d) Angle = π/2

2. Complex number on imaginary axis means:
   (a) Real part = 0
   (b) Imaginary part = 0

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(c) Modulus = 0

(d) Angle = 0

3. Complex conjugates are symmetric about:

(a) Real axis

(b) Imaginary axis

(c) Origin

(d) Line y = x

4. $-1 + i$ lies in:

(a) 1st quadrant

(b) 2nd quadrant

(c) 3rd quadrant

(d) 4th quadrant

5. All points equidistant from origin form a:

(a) Circle

(b) Line

(c) Triangle

(d) Parabola

7. Complex Numbers in Polar & Cartesian Forms

1. If $z = 1 + i$, then its modulus $r =$:

(a) $\sqrt{2}$

(b) 2

(c) 1

(d) $\sqrt{3}$

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2. The argument of $z = -1 + i$ is:
   (a) $\pi/4$
   (b) $3\pi/4$
   (c) $-\pi/4$
   (d) $-3\pi/4$

3. Convert polar form $z = 2e^{i\pi/3}$ to Cartesian:
   (a) $1 + i\sqrt{3}$
   (b) $-1 + i\sqrt{3}$
   (c) $2 + 2i$
   (d) $\sqrt{3} + i$

4. Polar to Cartesian conversion formulas are:
   (a) $x = r\cos\ \theta, y = r\sin\ \theta$
   (b) $r = x + y$
   (c) $\theta = xy$
   (d) None

5. In polar form, multiplication of complex numbers becomes:
   (a) Add moduli
   (b) Add angles
   (c) Multiply moduli and add angles
   (d) Subtract moduli

6. De Moivre's Theorem is useful for:
   (a) Raising complex number to power n
   (b) Finding conjugates
   (c) Rationalizing denominator
   (d) Integration

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7. Which identity is true?
   (a) $\cos^2\theta + \sin^2\theta = 0$
   (b) $|z| = r$
   (c) $z = r(\cos\theta - i\sin\theta)$
   (d) None

8. The polar form of $z = -1 - i$ is:
   (a) $\sqrt{2}e^{-3\pi/4}$
   (b) $\sqrt{2}e^{3\pi/4}$
   (c) $e^{-i\pi/2}$
   (d) $e^{i\pi/4}$

8. Complex Numbers in Encryption and Other Applications

1. In encryption, complex numbers are used in:
   (a) Public key cryptography
   (b) Graph theory
   (c) Modulus equations
   (d) Real analysis

2. Which complex operation is most helpful in rotation transformations?
   (a) Multiplication
   (b) Conjugation
   (c) Addition
   (d) Modulus

3. Complex roots are essential in solving:
   (a) Linear equations

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(b) Real inequalities

(c) Quadratics with negative discriminant

(d) Factorials

4. Complex numbers are used in signal processing to represent:

(a) Frequencies & phase

(b) Colour intensity

(c) Temperature

(d) Noise only

5. Which field uses complex numbers in wave representation?

(a) Biology

(b) Chemistry

(c) Physics

(d) Accounting

6. In encryption, complex numbers are useful because:

(a) They store large integers

(b) They rotate & scale data points

(c) They perform quick multiplication only

(d) They are irrational

7. Which is a benefit of complex numbers in AI?

(a) Efficient sorting

(b) Faster division

(c) Richer representations for neural networks

(d) Memory allocation

8. Fourier Transform uses complex numbers to:

(a) Compress images

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(b) Solve matrix inverses

(c) Analyze time-frequency signals

(d) Encrypt passwords only

9. In quantum computing, complex amplitudes help represent:

(a) Bits

(b) States

(c) Probabilities

(d) Real numbers

10. Which system uses Euler's formula in rotation matrices?

(a) Economics

(b) Robotics

(c) Education

(d) Management

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